

AI FOR SOCIAL GOOD: Social Media Analytics for Mental Health

Minor project report submitted in partial fulfillment of the requirement for the degree of
Bachelor of Technology

in

Computer Science and Engineering

By

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TABLE OF CONTENT

Title	Page No.
Certificate	I
Acknowledgement	II
Abstract	III
Chapter-1 (Introduction)	1
Chapter-2 (Feasibility Study, Requirements Analysis, and Design)	4
Chapter-3 (Implementation)	9
Chapter-4 (Results)	17
References	20

CERTIFICATE

We hereby certify that the work which is being presented in the project report titled “AI for Social Good” in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering and submitted to the Department of Computer Science And Engineering, Jaypee University of Information Technology, Waknaghat is an authentic record of our own work carried out during the period from January 2024 to May 2024 under the supervision of Dr. Aman Sharma, Department of Computer Science and Engineering, Jaypee University of Information Technology, Waknaghat.

The matter presented in this project report has not been submitted for the award of any other degree at this or any other university.

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This is to certify that the above statement made by the candidates is correct and true to the best of my knowledge.

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ABSTRACT

By carrying out our research through machine learning applications, the information about the social status of Twitter users and whether such individuals have any of the nine mental health disorders is being analyzed. Two datasets were created: it is the "observed" set consisting of 1,500 users who posted during a certain period where their condition was directly described as one of the mood disorders (e.g., depression, anxiety), g. , bipolar disorder) and depressive disorders (or depression), obsessive-compulsive disorders, Post-Traumatic Stress Disorder, schizophrenia, and ADHD; and the "control" set. Subsequently, two tasks were trained using conventional machine learning and deep learning classifiers: multiclassing for diagnosis and classifications into two classes diagnosed and non-diagnosed. We employed the strategy of obtaining a collection of tweets directly from the users who admit to having mental health problems and from those who do not self-report mental illness. Cleaning of data included removing the stop words and Twitter terms, converting text to lowercase formats, extracting the retweets, and applying stemming and part-of-speech tagging. The lexical features, parts of speech n-grams, topic modelling, LIWC analysis, and word embedding were extracted. Binary and multiclass classification tasks were done by using classifiers such as Naive Bayes, Logistic Regression, and Multilayer Perceptron. Data pre-processing in terms of detecting and classifying users as having or not having a mental disorder and a particular mental disorder was carried out. Privacy was resolved with the help of the Federated Learning model during the model training stage.

The evaluation of Logistic Regression and MLP models was based on AUC, F1 score, precision, and recall that confirm their classification performance. The results show that binary and multiclass classification tasks can successfully spot mental illness amongst tweets from English-speaking Twitter users, distinguishing between the diagnosed and undiagnosed cases. Naive Bayes and Logistic Regression were the most effective algorithms when it comes to precision, meaning the highest predictive accuracy.

These results show that, indeed, binary and multiclass classification can accurately categorize diagnosed and non-diagnosed users, and social networks are the source of data and the text analysis methods that are needed for mental health diagnosis, monitoring, and treatment assisting.

Chapter 01: INTRODUCTION

1.1 Introduction

It seems that digital technology, or, more precisely, social media is redefining the way we perceive mental health today and opens up a range of new opportunities for it to be timely diagnosed and treated [3]. This new technique, digital phenotyping, meshes the fields of data science, psychology, and behavioural study to dig into one's digital track and define certain features that can later be associated with mental health disorders[15,16].

Nevertheless, technological progress at the same time presents numerous challenges – such as ethical issues and data protection – which new regulations have to deal with. The process of collecting user data from social media platforms and their subsequent analysis requires an approach that handles the special information involved with consent and the ability for misuse[19]. The user bases of social media platforms, however, might not necessarily encompass the whole population one would like to analyze, which could lead to distortions in the collected data. Secondly, there is always the chance of misidentification, either over or under-diagnosis of people experiencing mental health problems which can have fatal effects[4, 7-9].

Despite such risks, the benefits of this being witnessed are substantial. Analyzing social networking patterns such as language, feelings, or activity of the users may turn out to be the precursor of mental health problems like depression or anxiety[17]. With the use of machine learning algorithms on this data set, it is possible to reveal integral behavioural traits that in a concealed manner may indicate the onset of mental health disorders. The application of the new mental health assessment approach may yield useful information to physicians on their patients' behavioural patterns in length, which provides an extra dimension to traditional diagnostic tools and could help doctors tailor treatments according to the patient's situation[9,10].

The interdisciplinary approach of this science transpires to be the major factor moving the discipline forward. Firstly, data scientists provide their ability to understand machine learning models and computational analysis techniques[22]. Further, psychologists and clinicians bring their knowledge regarding mental health problems and methods of diagnosis. Ethicists explore and implement a paradigm that would guarantee that the research is conducted with due respect to privacy and consent, which will, in turn, develop a healthy relationship between people and mental-tech care machines[29-31].

Still, unanswered questions need to be addressed such as: How can we set reliable ethical standards for digital profiling and make the models transparent? Social media data analysis also requires working towards inclusive and culturally sensitive methods[18]. Even, some social media behaviour is different across various demographic groups. The implementation of ethics-based practices and a teamwork approach in digital phenotyping would have the potential to advance early diagnostics and intervention in psychiatry[34,35].

1.2 Objective

The primary objectives of this project are:

- Perform binary classification of tweets (e.g., diagnosed vs. controlled).
- Perform multiclass classification of tweets into detailed sentiment categories.
- Implement and assess federated learning methodologies within the sentiment classification framework.

1.3 Motivation

The motivation behind exploring digital phenotyping in mental health stems from the escalating global burden of mental disorders and the urgent need for innovative approaches to early detection and intervention. Mental health issues like depression and anxiety are among the leading causes of disability worldwide, yet they often go undetected or untreated due to factors such as stigma, lack of awareness, and inadequate access to mental health services[1-3].

Digital technology, particularly social media, offers a unique opportunity to address these challenges. As people increasingly use platforms like Twitter and Facebook to express themselves and interact with others, their online behaviour creates a digital footprint that can reveal patterns indicative of mental health status. By analyzing these digital footprints, researchers can leverage data science techniques such as machine learning and natural language processing to identify early signs of mental health disorders[7].

The advantages of this approach are manifold. Firstly, it allows for continuous, real-time monitoring of mental health, which can lead to earlier detection and intervention compared to traditional methods that rely on self-reporting or infrequent clinical visits. Secondly, it reduces the barriers associated with stigma and access, as it passively collects data without requiring individuals to actively seek help. Thirdly, it can provide insights at a population level, helping to identify trends and risk factors that might be invisible through conventional epidemiological methods.

Our inspiration for this project came from a research paper [11] that analyzed Twitter datasets for English and Spanish-speaking users to perform sentiment analysis. This study demonstrated the potential of social media data to reflect users' mental health states through their expressed sentiments. Building on these findings, we aimed to classify tweets according to sentiment, thereby facilitating a deeper understanding of the mental health landscape reflected in social media data. Additionally, we explored the use of federated learning methodologies to enhance the privacy and security of the data analysis process.

1.4 Language Used

The programming language used in this project is Python, along with the various other libraries, given its popularity in the field of machine learning and natural language processing.

1.5 Technical Requirements (Hardware)

- Google Colab connectivity with T4-GPU enabled

For the application interface

- RAM: 8GB
- Storage: 2-3 GB Free space
- CPU: 1.6GHz or faster
- Architecture: 32-bit or 64-bit
- Operating System: Windows 10 and above

1.6 Deliverables/Outcomes

These outcomes and deliverables collectively contribute to advancing our understanding of the relationship between social media behaviour and mental health, to improve early detection, intervention, and support for individuals experiencing mental health challenges.

1. Trained Machine Learning Models:

- Deliver high-performing machine learning models trained on social media data to classify users into diagnosed and non-diagnosed groups and identify specific mental health disorders.

2. Classification Reports and Metrics:

- Provide detailed reports on model performance, including evaluation metrics such as accuracy, precision, recall, F1 score, and area under the ROC curve (AUC).

3. Feature Extraction Framework:

- Develop a framework for extracting relevant features from social media text data, including lexical, syntactic, semantic, and sentiment-based features.

4. Data Preprocessing Pipeline:

- Create a robust data preprocessing pipeline to clean and prepare social media data for analysis, including steps such as removing retweets, stop word removal, and part-of-speech tagging.

5. Privacy-Preserving Techniques:

- Implement privacy-preserving techniques such as federated learning to address concerns around data privacy and confidentiality while training machine learning models.

6. Research Paper or Report:

- Produce a research paper or report summarizing the methodology, findings, limitations, and implications of the study for publication in academic journals or presentations at conferences.

7. Ethical Guidelines and Recommendations:

- Offer ethical guidelines and recommendations for the responsible use of social media data in mental health research, including considerations for informed consent, data anonymization, and user privacy.

8. Educational Resources and Outreach:

- Create educational resources and outreach materials to raise awareness about the project's findings, promote mental health literacy, and encourage responsible use of social media platforms.

Chapter 02: Feasibility Study, Requirements Analysis and Design

2.1 Feasibility Study

2.1.1 Problem Definition

The prevalence of mental health disorders is a growing concern worldwide, with conditions such as depression, anxiety, bipolar disorder, and PTSD affecting millions of individuals. Despite the significant impact of these disorders, early detection and intervention remain challenging due to factors like stigma, lack of awareness, and limited access to mental health services. Traditional diagnostic methods often rely on self-reporting and clinical assessments, which can be time-consuming and prone to underreporting.

In parallel, the rise of social media platforms like Twitter has led to an unprecedented amount of publicly available user-generated content. This content often contains valuable insights into users' thoughts, feelings, and behaviours, making it a potential resource for identifying mental health issues. However, analyzing this vast amount of unstructured data to detect mental health disorders is complex and requires sophisticated machine-learning techniques.

2.1.2 Problem Analysis

1. Data Collection and Reliability:

- **Self-Reported Data:** The reliance on self-reported data on social media may lead to inaccuracies, as users might misrepresent their mental health status or fail to disclose it accurately.
- **Data Volume and Variety:** Social media data is vast and diverse, making it challenging to filter relevant information and manage data effectively.

2. Privacy and Ethical Concerns:

- **User Consent:** Users may not be aware that their public posts are being used for research, raising ethical concerns regarding consent and data usage.
- **Data Confidentiality:** Handling sensitive mental health data necessitates robust measures to ensure user privacy and data protection.

3. Technical Challenges in Machine Learning:

- **Feature Extraction:** Extracting meaningful features from unstructured text data, such as tweets, requires sophisticated natural language processing techniques.
- **Model Selection:** Identifying and implementing appropriate machine learning models that can accurately classify mental health statuses based on social media data.
- **Performance and Scalability:** Ensuring that the system can process and analyze large volumes of data efficiently to provide timely insights.

2.1.3 Literature Survey

Table 1. An overview of a few studies on the identification of mental disorders via social media.

S.No.	Year and Author	Source	Attributes	Algorithms	Outcomes
1	Jamil et al. 2017 [4]	Twitter (CLPsych & BellLetsTalk)	Depression word count, pronoun counts, LIWC, NRC sentiment lexicon, emoticon frequency.	Linear SVM	F1: 0.66 P: 0.58 R: 0.77 Acc: 0.61 (Depression User-level)
2	Cohan et al 2018. [5]	Reddit (SMHD)	TF-IDF vectorization	Logistic Regression, SVM, XGboost, FastText, and CNN	F1: 0.27 P: 0.23 R: 0.48 (Multi-label)
3	Husseini et al. 2018 [17]	Twitter (CLPsych)	Word pre- trained on the CLPsych 2015 Shared task data	Logistic Regression, CNN with word embeddings(Gl oVe, fastText, and own corpora)	AUC:0.95 F1: 0.87 P: 0.87 R: 0.87 Acc: 0.88 (CLPsych 2015)
4	Husseini et al. 2018 [17]	Twitter (BellLets Talk)	Word pre- trained on the CLPsych 2015	Logistic Regression, CNN with	AUC:0.92 F1: 0.82 P: 0.82

			Shared task data	word embeddings(GloVe, fastText and own corpora)	R: 0.84 Acc: 0.83 (BellLetsTalk)
5	Tadesse et al 2019 [6]	Reddit	LIWC, LDA, n-grams, (TF-IDF)	Logistic Regression, SVM, Random Forest, Adaptive Boosting and Multilayer perceptron	F1: 0.93 P: 0.90 R: 0.92 Acc: 0.91 (Depression)
6	Aguilera et al. 2020 [13]	Reddit (eRisk)	TF-IDF and word-embeddings (GloVe, fast Text, Word2vec	Baseline: ocSVM, ocKNN, SVM, KNN.	F1: 0.63 (Depression) F1: 0.81 (Anorexia)
7	Malviya et al. 2021 [18]	Reddit	Word2vec, TF-IDF	Linear and Boosting classifiers, and Transformers)	F1: 0.98 Acc: 0.98 (Depression)
8	Uban et al 2021 [23]	Reddit (eRisk)	GloVe embeddings, content, style features, LIWC, emotions and sentiments	Logistic Regression, BiLSTM with attention, HAN, CNN, and Transformers(RoBERTa and	AUC:0.87 F1: 0.65 (Self-harm) AUC:0.96 FI: 0.61 (Anorexia) AUC:0.83 FI: 0.45

				Albert)	(Depression)
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2.1.4 Solution

To address these challenges, we propose a comprehensive framework that leverages machine learning and natural language processing to analyze Twitter data for mental health diagnostics. This involves:

- 1. Data Collection: Extracting tweets from users who self-report mental health conditions and a control group.
- 2. Data Preprocessing: Cleaning and standardizing the data to remove noise and irrelevant content.
- 3. Feature Extraction: Utilizing advanced techniques to extract meaningful features from text data.
- 4. Model Training: Training binary and multiclass classifiers using algorithms like Naive Bayes, Logistic Regression, and MLP.
- 5. Federated Learning: Ensuring privacy-preserving training by distributing the model training process.
- 6. Evaluation: Using comprehensive metrics to evaluate and validate model performance.

By integrating these components, the project aims to provide a scalable, accurate, and privacy-preserving solution for early detection and monitoring of mental health disorders, ultimately contributing to improved mental health care and intervention strategies.

2.2 Requirements

2.2.1 Functional Requirements

- 1. Data Collection: The system must collect tweets from users who have self-reported mental health conditions and from control users who have not disclosed any such conditions.
- 2. Data Preprocessing: The system must preprocess the text data by removing retweets, converting text to lowercase, and removing stop words, special characters, digits, and Twitter-specific elements (hashtags, mentions, URLs).
- 3. Feature Extraction: Extract word n-grams, and character q-grams, and calculate TF-IDF values. Utilize pre-trained word embeddings like GloVe and fastText.
- 4. Model Evaluation: The system must evaluate model performance using metrics like AUC, F1 score, precision, and recall.
- 5. User Interface: Provides a user-friendly dashboard to display results, including performance metrics and visualizations like confusion matrices.

2.2.2 Non-Functional Requirements

1. Performance: The system should handle large volumes of tweet data efficiently.
2. Reliability: The system should handle noisy and incomplete data effectively without significant performance degradation.
3. Usability: The interface should be intuitive and easy to navigate for users with varying levels of technical expertise.
4. Security: Ensure the privacy of user data, especially sensitive information related to mental health.
5. Maintainability: Implement robust error handling and logging mechanisms to troubleshoot and resolve issues efficiently.

2.3 E-R Diagram / Data-Flow Diagram (DFD)

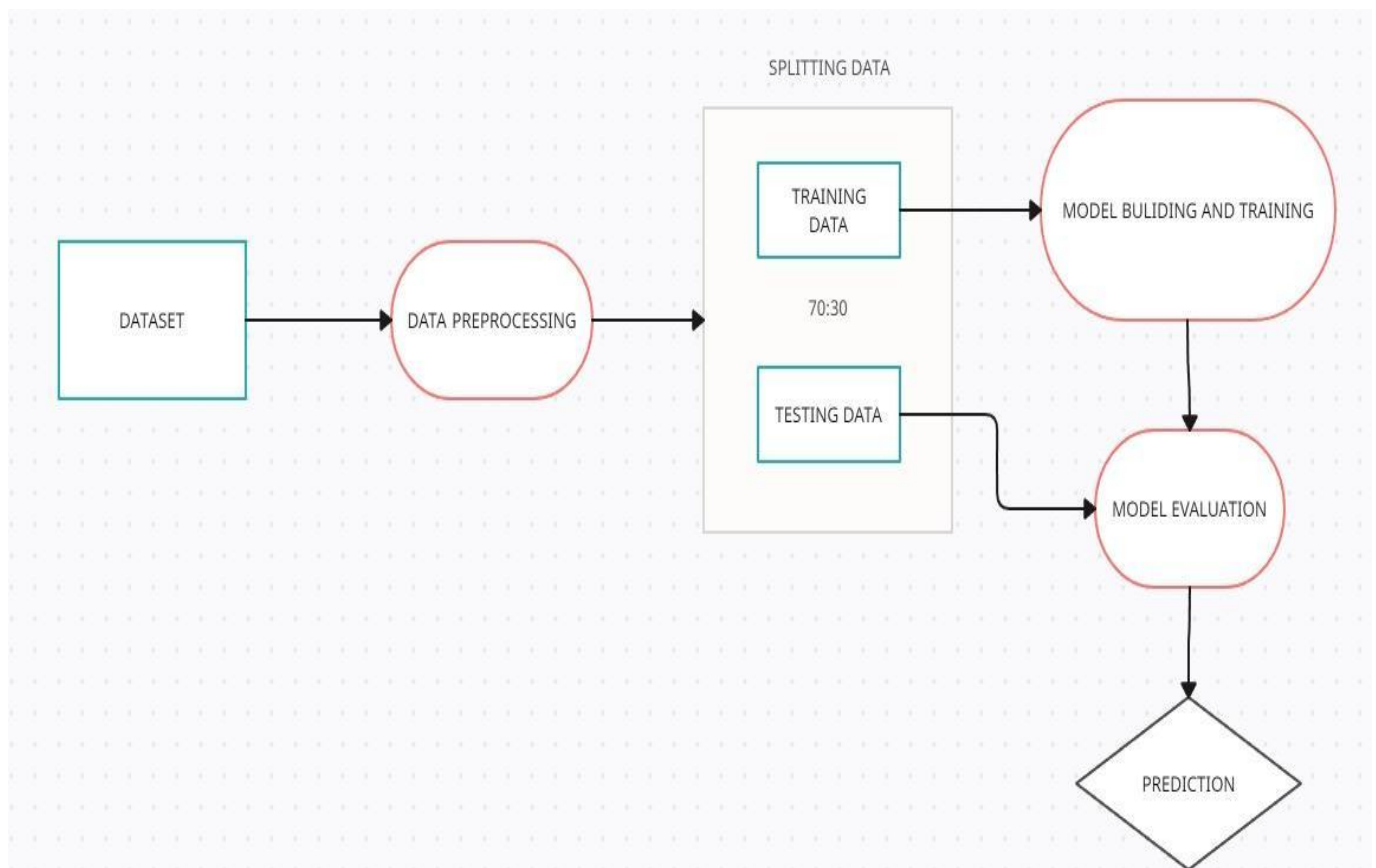


Fig 1. Flow Chart

Chapter 03: IMPLEMENTATION

3.1 Data Set Used in the Minor Project

The dataset used in this project is a collection of Twitter posts labeled with information about the user's mental health status. The text data (post_text) is used as the feature for classification, and the label column provides the target variable for binary classification. The data is preprocessed by cleaning, encoding, and splitting into training and testing sets to facilitate model training and evaluation.

3.2 Data Set Features

3.2.1 Types of Data Set

The dataset used in this project is a labeled Twitter dataset containing posts (tweets) from users. The dataset has the following characteristics:

1. Text Data: The post_text column contains the textual content of the tweets. This is the primary feature used for classification.
2. Class Labels: The label column indicates whether the user is diagnosed with a mental health condition or not. There are two possible labels: "diagnosed" and "not diagnosed", which are encoded numerically (0 for diagnosed, 1 for not diagnosed).

3.2.2 Number of Attributes, fields, and description of the data set

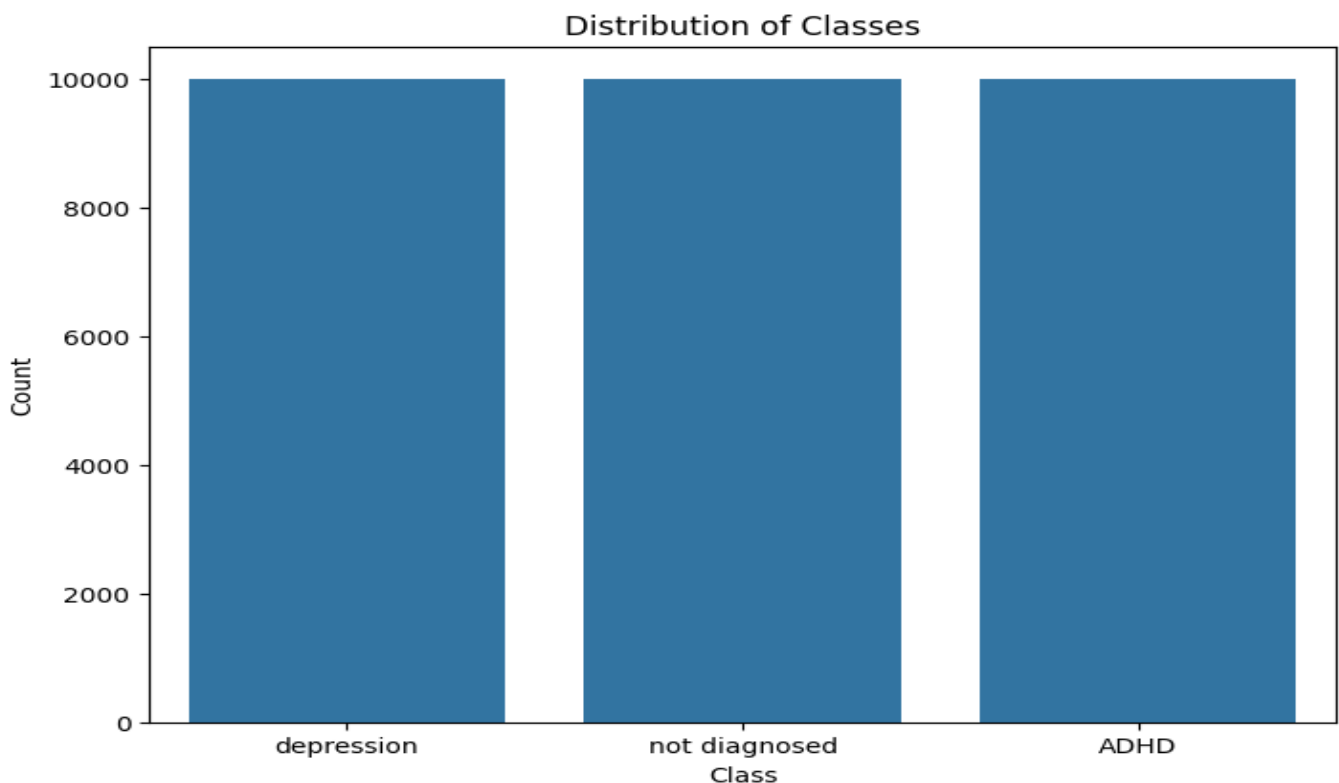


Fig 2. Attributes used in the dataset with their counts

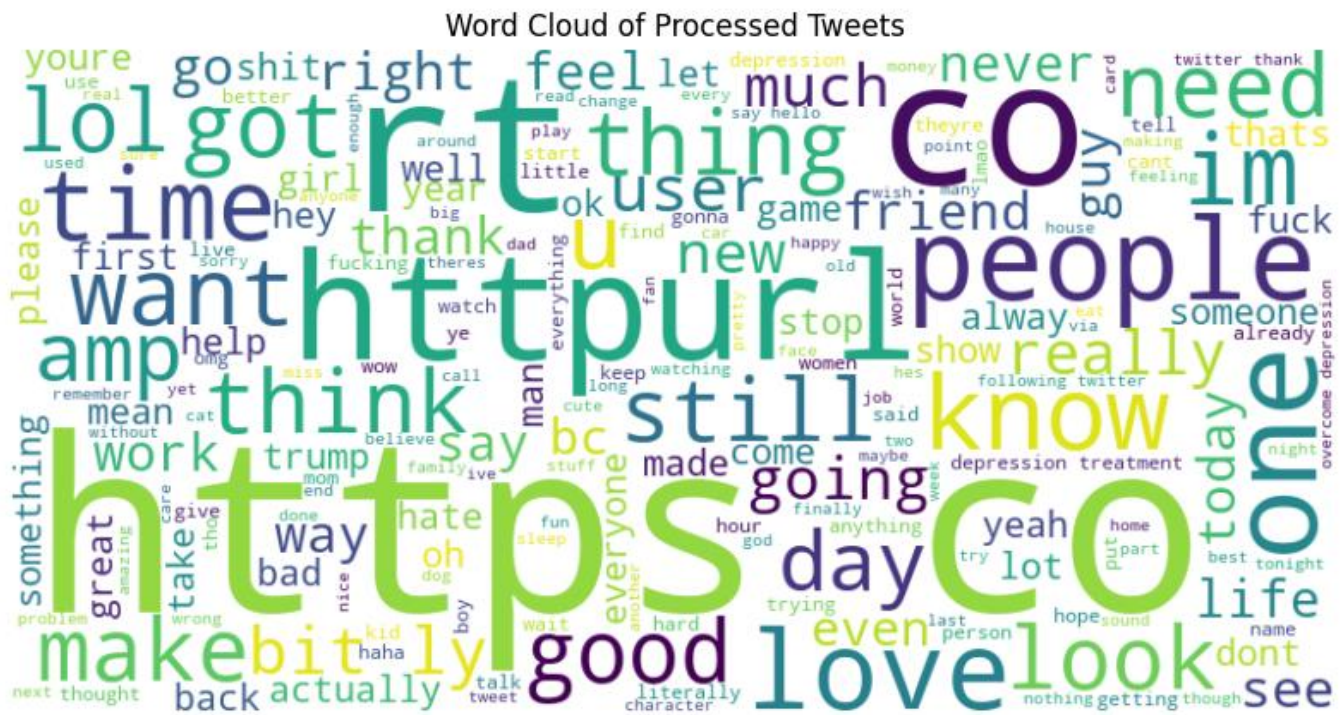


Fig 3. Word Cloud of Processed Tweets.

3.3 Algorithm / Pseudo code of the Project Problem

```
def build_mlp_model(input_shape, num_classes):
    model = tf.keras.Sequential([
        tf.keras.layers.Dense(128, activation='relu', input_shape=(input_shape,)),
        tf.keras.layers.Dropout(0.5),
        tf.keras.layers.Dense(64, activation='relu'),
        tf.keras.layers.Dropout(0.5),
        tf.keras.layers.Dense(num_classes, activation='softmax')
    ])
    return model

num_global_rounds = 50
local_epochs = 1
batch_size = 32
input_shape = X_train.shape[1]
num_classes = y_train.shape[1]

global_model = build_mlp_model(input_shape, num_classes)

global_model.compile(
    optimizer=tf.keras.optimizers.SGD(learning_rate=0.01),
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

```

global_weights = global_model.get_weights()

test accuracies = []

for round in range(num_global_rounds):
    scaled_local_weights = []

    for client_name, (client_X, client_y) in clients_data.items():
        client_model = build_mlp_model(input_shape, num_classes)

        client_model.compile(
            optimizer=tf.keras.optimizers.SGD(learning_rate=0.01),
            loss="categorical_crossentropy",
            metrics=["accuracy"]
        )

        client_model.set_weights(global_weights)

        client_model.fit(client_X, client_y, batch_size=batch_size,
epochs=local_epochs, verbose=0)

        new_weights = client_model.get_weights()

        scaling_factor = weight_scaling_factor(client_X, X_train)
        scaled_weights = scale_model_weights(new_weights, scaling_factor)

        scaled_local_weights.append(scaled_weights)

    global_weights = sum_scaled_weights(scaled_local_weights)
    global_model.set_weights(global_weights)

    test_loss, test_accuracy = global_model.evaluate(X_test, y_test, verbose=0)
    test accuracies.append(test_accuracy)
    print(f"Round {round + 1}: Test Accuracy = {test_accuracy * 100:.2f}%")

print("Test Accuracies for each communication round:", test accuracies

```

3.4 Flow graph of the Federated Learning Model

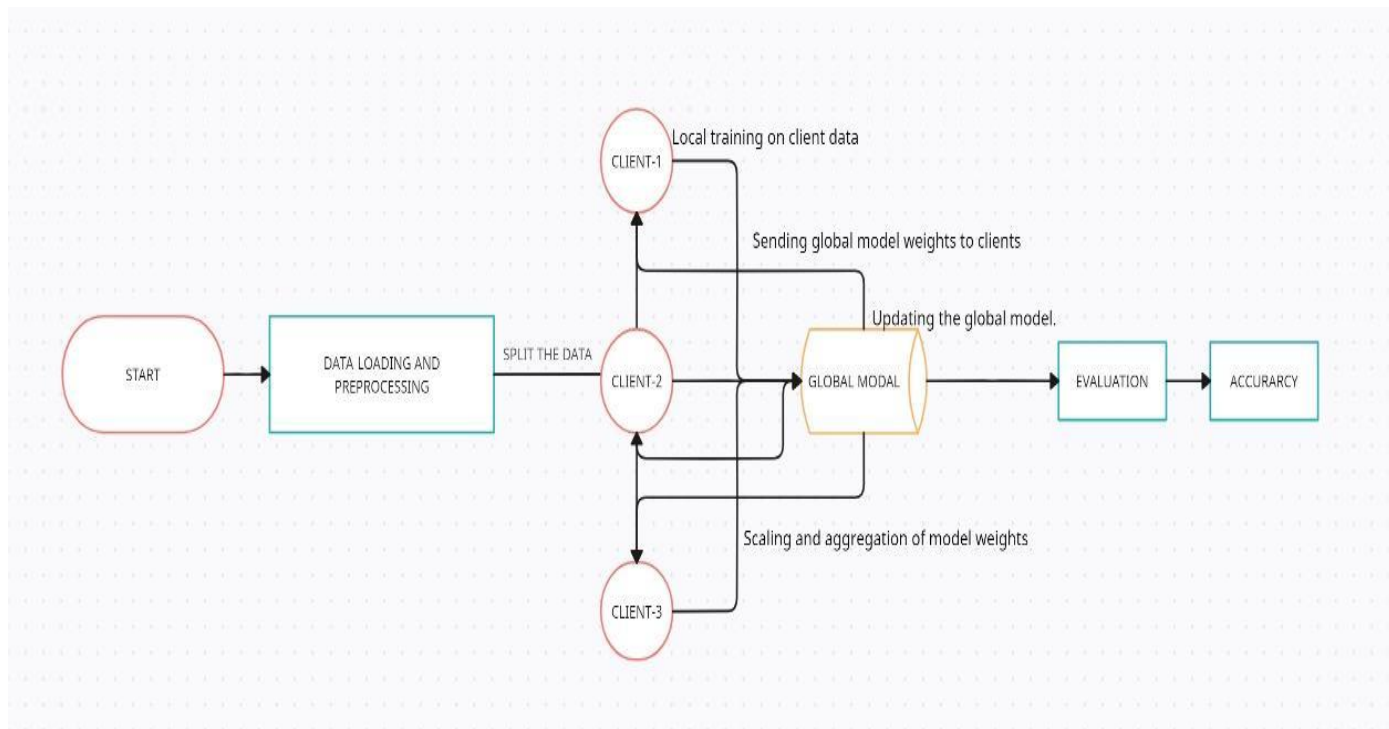


Fig 4. Flow graph depicting the working of Federated Learning Model.

3.5 Screenshots of various stages of the project

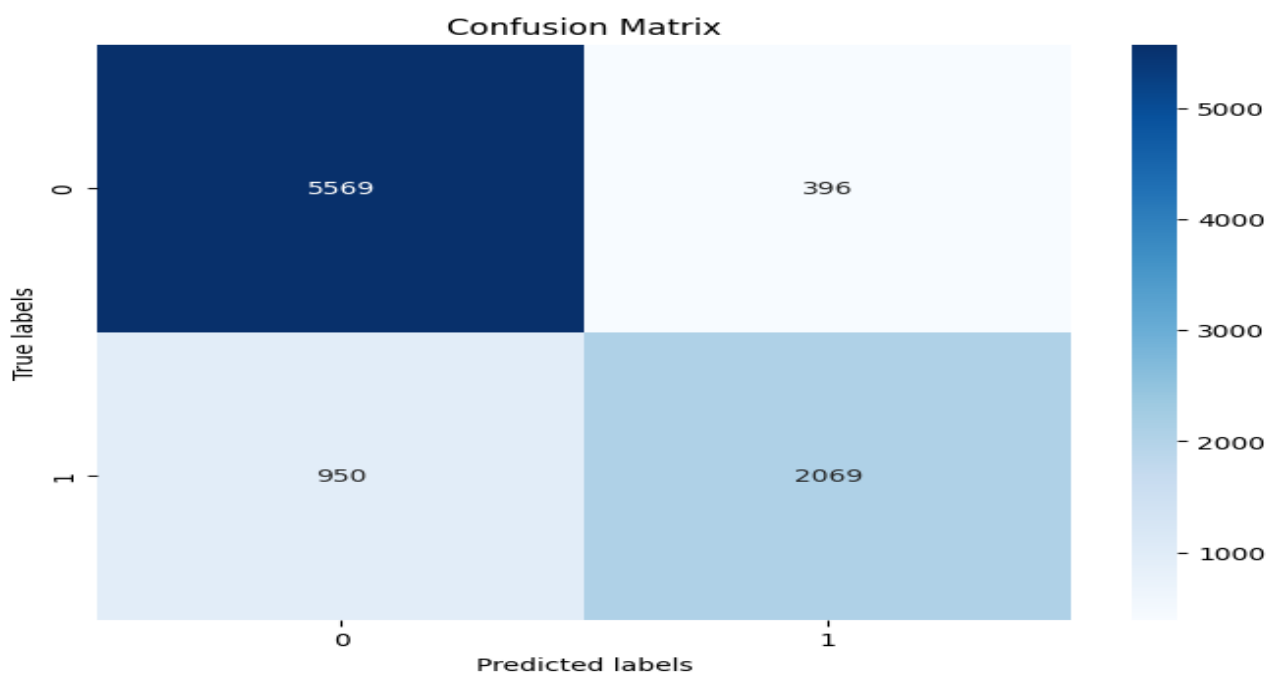


Fig 5. Confusion Matrix for MLP

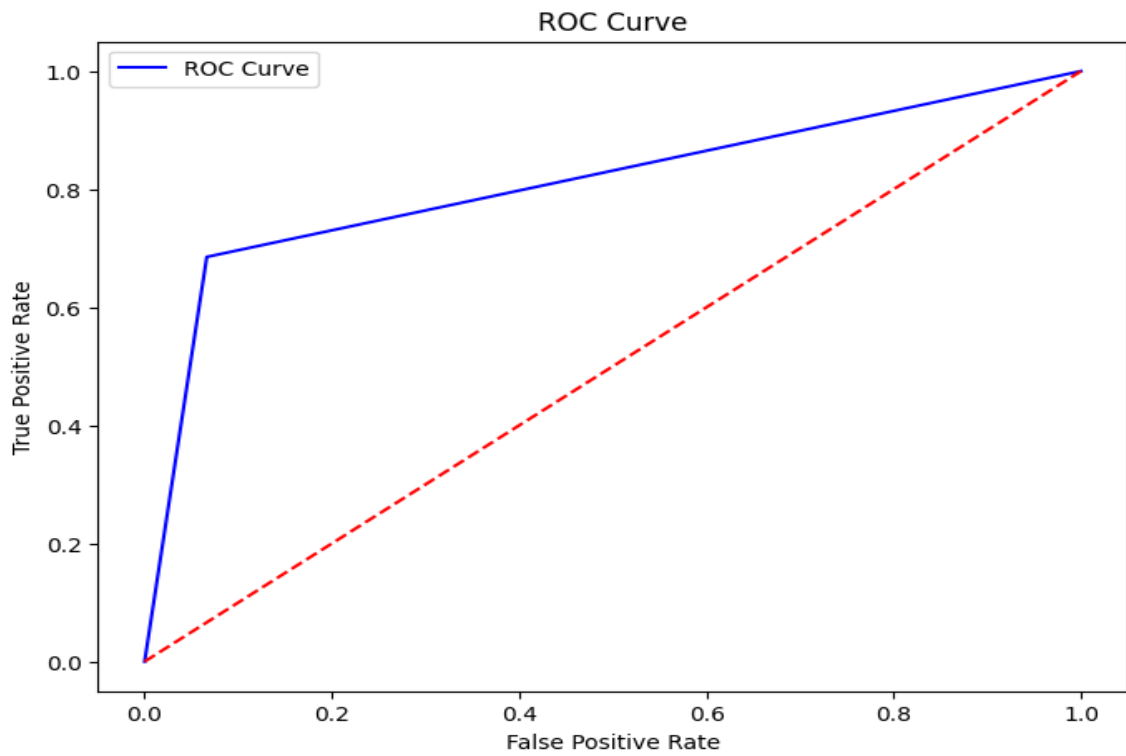


Fig 6. ROC Curve obtained after applying MLP

MLP with TfidfVectorizer

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.94	0.89	5965
1	0.85	0.67	0.75	3019
accuracy			0.85	8984
macro avg	0.85	0.80	0.82	8984
weighted avg	0.85	0.85	0.84	8984

Fig 7. Classification Report for MLP with TF-IDF

Chapter 04: RESULTS

4.1 Discussion on the Results Achieved

The project involved training multiple machine learning models using two different text vectorization techniques (Count Vectorizer and Tfidf Vectorizer) to classify tweets into two categories: "diagnosed" (users diagnosed with a mental health condition) and "not diagnosed" (users not diagnosed with a mental health condition).

The MLP Classifier with Count Vectorizer and Logistic Regression with Count Vectorizer both achieved an overall accuracy of 85%, providing a good balance between precision and recall for both classes. Naive Bayes models showed high precision but lower recall for the "not diagnosed" class, indicating they are better at identifying "diagnosed" users but tend to miss some "not diagnosed" cases.

Tf-idf Vectorizer tends to improve precision but can lead to lower recall for the minority class, resulting in more false negatives.

Table 2. Comparison between Binary classification and Multiclass classification

MODELS	BINARY CLASSIFICATION	MULTICLASS CLASSIFICATION
Federated Learning	0.77	0.639
Naive Bayes	0.84	0.77
MLP	0.85	0.77
Logistic Regression	0.85	0.76

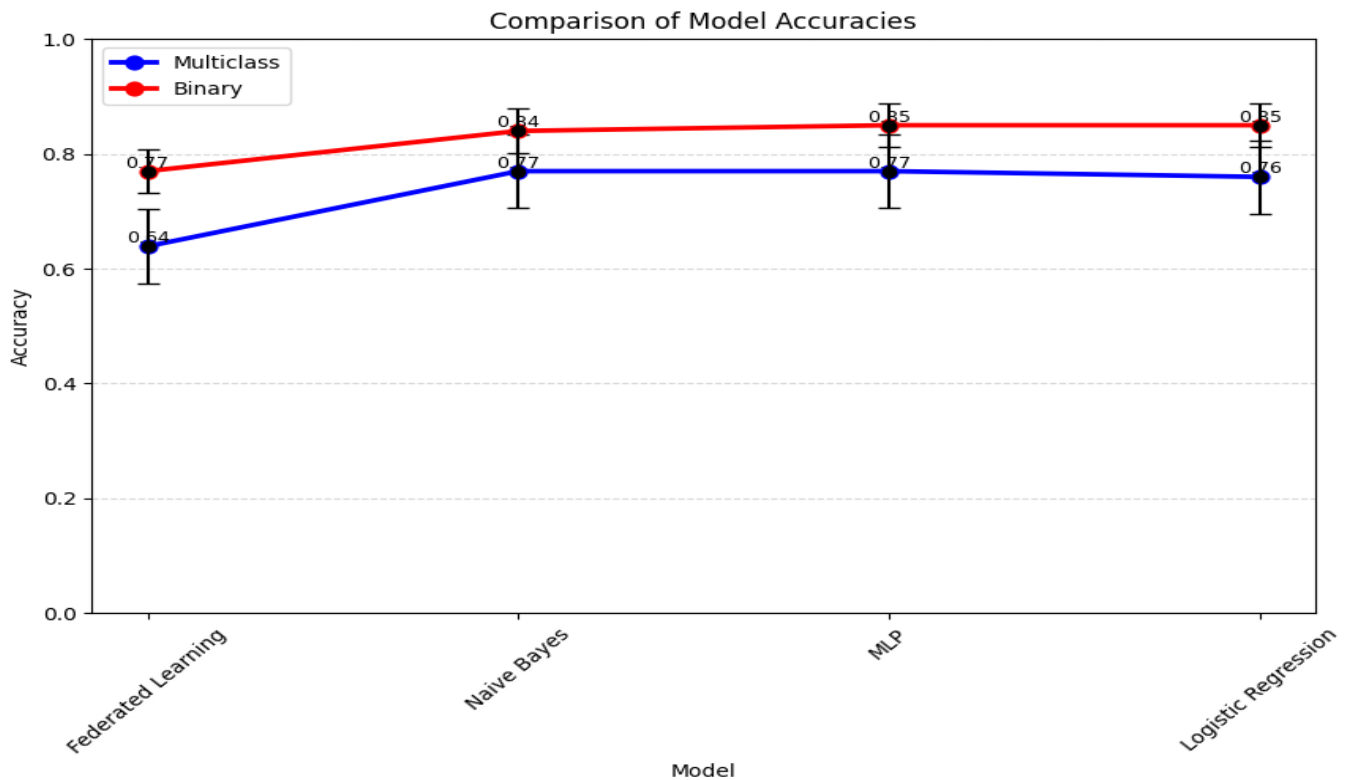


Fig 8. Comparison between accuracies

4.2 Application of the Minor Project

Below are the applications of the project:

- **Proactive Identification:** By using machine learning models to analyse social media behaviour to detect early indicators of mental health illnesses, timely intervention can be made possible.
- **Diagnostic Support:** By offering more data-driven insights, help mental health practitioners diagnose disorders like depression, anxiety, PTSD, and others.
- **Mental Health Monitoring: Continuous Monitoring:** Put in place systems to keep an eye on people's mental health conditions over time and send out notifications in real-time if there are any notable changes or possible emergencies.
- **Analyse trends and patterns in mental health over extended periods** to help patients and healthcare professionals comprehend how illnesses develop.
- **Tailored Recommendations:** Provide individualised mental health resources, coping mechanisms, and assistance according to the models' identification of particular illnesses and user behaviour.
- **Behavioural input:** Give users input regarding their state of mental health, promoting self-care and proactive management.
- **Assistance for Counsellors and Therapists:** Provide mental health providers with

more information and tools so they can better comprehend patients' situations and adjust treatment plans.

- **Resource Allocation:** By identifying regions or populations with greater needs, you can help mental health organisations allocate resources more wisely.
- **Academic Research:** Offer scholars examining the relationship between language use, mental health, and social media a useful dataset and analytical methodology.
- **Cross-Disciplinary Studies:** Promote multidisciplinary investigations into new facets of mental health by bringing together data science, social sciences, and psychology.

4.3 Limitation of the Minor Project

- **Selection Bias:** Because the dataset is based on people who freely report their mental health condition on Twitter, it may not fully represent the broader population.
- **Demographic Bias:** There could be biases in the analysis and restricted generalizability of the results due to the potential lack of demographic diversity among Twitter users.
- **incorrectly classified. Accuracy of Diagnosis:** The self-reported mental health diagnoses made by users may not always be precise or supported by clinical evidence, which could cause the dataset to be
- **Underreporting or Stigma:** People may be reluctant to talk about their mental health issues out of fear of being stigmatised or out of privacy concerns, which causes some disorders to be underrepresented.
- **Overfitting:** When machine learning models overfit to the training set, they perform poorly on new data or in practical settings.
- **Model Generalisation:** The applicability of models trained on a single dataset or social media platform may be limited in a variety of settings due to their poor generalisation to other platforms or demographics.

4.4 Future Work

By addressing these areas in future work, the project can significantly enhance its impact, accuracy, and usability, ultimately contributing to better mental health outcomes on a larger scale.

1. Improvement in Model Accuracy and Robustness:

- **Advanced NLP Techniques:** Incorporate more sophisticated natural language processing (NLP) models, such as transformers (e.g., BERT, GPT), to improve the accuracy and robustness of mental health disorder classification.
- **Multimodal Data Analysis:** Extend the analysis to include other types of data, such as images, videos, and user interactions, to capture a more holistic view of the user's mental state.

2. Expansion to Other Social Media Platforms:

- Cross-Platform Analysis: Extend the framework to include data from other social media platforms like Facebook, Instagram, Reddit, and TikTok to improve generalizability and reach a broader audience.
- Platform-Specific Models: Develop and optimize models specifically tailored to the unique characteristics and user behaviour of each social media platform.

3. Integration with Health Care Systems:

- Real-Time Monitoring Tools: Develop real-time monitoring tools that can alert healthcare providers or users themselves about potential mental health issues, facilitating timely intervention.
- Collaborative Platforms: Create platforms for collaboration between data scientists, mental health professionals, and policymakers to ensure that the insights generated are actionable and beneficial for public health.
-

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